FASE: Decentralized Compute and Market Structure

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Agreements

The following sentences stand on their own. These are candidate agreements and reported points of convergence for working-group review. Inclusion does not establish consensus, verify a technical claim, or record formal adoption. The working group must confirm each decision before it is cited as such. Claims that were only discussed, including figures and alternatives that were not adopted, remain in full under Discussion.

Definitions will determine how compute is regulated. Fungibility, price discovery, settlement, storage and inventory, counterparties, and recognition are the dimensions the short paper uses. A "mature blockchain" test was set aside.

Under US law almost everything can be a commodity if a future is dealt in it. The CFTC regulates futures and swaps, not the spot market. A data center selling spare capacity, or an edge network selling spot compute, is not a CFTC-regulated entity merely for selling the service.

The CME's H100 and B200 contracts are financially settled to a rental index, on a monthly average, which is a deliberate anti-manipulation and hedge-fit choice. They went through a formal CFTC process rather than self-certification. Data quality of privately reported rental prices is the supervisor's real concern. Trading is expected once the November review window closes.

A viable contract needs hedgers, both sides of the service, speculators who are not manipulators, real volatility, and trust in price and settlement.

Compute is a service that cannot be stored. It is closer to electricity than to oil, without electricity's grid-collapse shortage cost. Overcapacity should push prices toward incremental cost. Hardware generations should be handled by new contracts and by price-based rolls, the way option strikes are adjusted after a split, not by pretending old chips remain the same deliverable.

Most commodities already trade in decentralized over-the-counter markets. Bitcoin hashing was the compute example. Grades and multiple contracts are normal. Edge-device baskets should not be forced into one H100-equivalent without a published conversion.

Decentralized compute is unusual inside crypto because the paying customer can be outside crypto. Visibility of on-chain settlement is higher than visibility of cloud contracts, which is both a supervisory opportunity and a pseudonymity problem.

Real economic output is the test that separates a network from a circular token system: external buyers, real work product, stablecoin settlement. Fixed emissions that dwarf revenue are a subsidy and a distortion. Experimentation can fail; survival requires someone outside the system paying for the service.

The spot compute business is not a natural object of financial-market regulation. A futures venue is. Entering a regulated market without knowing how regulators work is the practical hurdle, not the commodity label by itself.

Indexes need verifiable, broad data. LIBOR is the model of how a surveyed index becomes a systemic price and then a scandal. A monthly average raises the cost of manipulation.

On-chain prices can inform the wider market if they are a statistically meaningful sample. They do not need to be the whole market.

Compute futures are valuable to equities as information about future demand, in either direction. They are not, by themselves, a forecast that the AI equity rally extends or breaks.

A chain can organize this market and is not the only technology that can. Osaka rice futures are the reminder that derivatives predate blockchains.

Venues that are already live should be grandfathered if they are not fraudulent. Rules should match the specific risk of the technology. The aims are no fraud, no disappearing customer funds, economic usefulness, and no systemic harm. Paperwork violations are not the point. Supervisors will lag the technology; monitoring, and writing rules after a real failure is visible, is the method that was recommended.

The historical playbook does not change because tokenization came first. Products exist, problems appear, regulation follows, and who pays for it is a political fight.

The audiences for the work are the crypto industry, decentralized-compute operators, and regulators, in the United States and elsewhere. The next writing should be chosen for those readers. Education, risk, and how to regulate an industry that already exists are the useful contents for regulators.

Documents considered

The session referred to the following documents, repositories, and standards. Bibliographic identifications and factual checks added during review are marked as editorial clarifications; they do not imply that the links were spoken or that the meeting adopted the findings. Names of the people who introduced the material are omitted.

Editorial identification: Compute-as-Commodity: Observable Flows in Decentralised Compute Infras-

tructure, draft study report linked by the session programme. Figures in this document retain their own dates and denominators; they are not substituted for unmatched session statistics.

Short BGIN paper on compute as a commodity and on centralized and decentralized compute market structure. Already published, in the session's account. No title spoken.

Longer companion report, not finished [unverified caption name]. About three to four weeks from a comment or publication step.

Session participation note, not yet posted, to go on Discourse. A study report of this session to be circulated for correction. Slides from both briefings.

CME H100 and B200 financially settled compute futures; Silicon Data H100 rental index, cited around $2.81 per GPU-hour; CFTC self-certification path (Bitcoin futures) versus the formal 45-plus-45-day path ending 9 November.

Networks discussed: Akash, Bittensor, Render, Aethir. Plume as a real-world-asset chain comparison. Hyperscalers and cloud share: AWS, Azure, Google, and Oracle among those named in the capex sentence.

Historical and legal references: Commodity Exchange Act; onion-futures ban; motion-picture box-office exclusion; California electricity market in 2000; Texas electricity stress; LIBOR; Osaka rice futures; New York taxi partition versus cashless ride services; self-certification of event contracts on a GPU-hour price.

A finished stablecoin paper was mentioned in the previous session without a title. MiCA 2.0 was named only as the afternoon topic.

Discussion

The following is the substance of the session. It preserves claims, figures, alternatives, and objections. Discussion does not establish a decision. Any candidate agreement requires explicit working-group confirmation.

Decentralized compute and market structure. A short BGIN paper on compute as a commodity, covering centralized and decentralized compute markets, was the base text. A longer companion report was described as not finished. The session paired a market-structure briefing on US compute futures with a walk through on-chain compute networks, then a question series. Several prepared questions were skipped for time.

Why this market is being treated as new. The interest in the short paper is that compute may be the first thing that is not yet fully defined as a commodity, already exists on-chain, and may then be defined as a commodity off-chain, forcing a look backward at what is already live. Decentralized compute was also described as one of the few blockchain business models that is economically circular: it can bring in real clients from outside crypto, and it can use the chain to widen access to compute, including edge devices. The session order was a market-structure briefing first, then the on-chain networks, then questions. A session note was not prepared in time and is to be posted on Discourse.

Definitions decide the regulation. The study defines a commodity market by fungibility, price

discovery, settlement, storage and inventory dynamics, counterparties, and regulatory recognition. It describes how several compute-as-commodity venues fit those dimensions. Definitions matter because they change how people think and how regulators act. US crypto-bill fights over the definition of a "mature blockchain" were set aside as the wrong cut; they were not taken up.

Under the Commodity Exchange Act, almost everything is a commodity: anything in which contracts for future delivery are or will be dealt in. Onions are excluded, because onion farmers were being harmed by futures and futures on onions are banned. Motion-picture box-office contracts are excluded because that industry did not want futures to signal that a film would fail. The logical endpoint of the definition is that a future can make a commodity out of something that is not a physical good at all.

CME compute futures. The CME has announced compute futures. The intended start was 5 October. That date had already passed at the time of the session. Two contracts: one on an H100 rental index for Nvidia Hopper H100 GPUs, sourced from Silicon Data [unverified: captions also say "some kind data"]; one on Nvidia Blackwell B200 GPUs. Both are financially settled. The exchange does not take on proving physical delivery; settlement is to the index. A recent print of the H100 rental index was $2.81 per GPU-hour. Settlement uses the average price for the month. A monthly average is harder to manipulate, because the price has to be moved on every day of the month, and it is closer to the exposure a hedger actually faces.

How a US futures contract gets listed. Two paths. Self-certification: the exchange notifies the CFTC and the contract proceeds unless the CFTC later objects. Bitcoin futures were listed that way, at the CME, and the CFTC allowed them. The other path is a formal request. The CFTC has 45 days to approve, and can add up to 45 days if the contract raises novel questions. For these compute contracts the second 45-day period runs out on 9 November. Approval, disapproval, or silence all resolve the request; silence is automatic approval. These contracts were described as having gone through the full request, not self-certification. Trading was expected shortly after 9 November. A later remark hoped for something before 5 November. Those two dates were not reconciled.

What the CFTC cares about is manipulation, and, more sharply here, data quality. The indexes are built from privately reported prices. GPU rentals are not formed in a transparent open-outcry market; they are private deals. Staff were expected to be asking where the prices come from and whether the indexes reflect the rental market.

What a viable futures market needs. Enough interest and enough players. Hedgers who have a real economic exposure and want to lay it off. Preferably both suppliers and demanders of the service. Existing compute demand and existing rental markets were treated as enough to check that box. Speculators, who provide liquidity by trading on price alone, bear risk the way an insurer does, and bring information into the price. They were distinguished from manipulators. The contract is useless without volatility, because a future is a risk-transfer product. Above those, trust: that prices are honest, that settlement works, and that the price tracks the exposure being hedged or speculated on.

Who becomes regulated. A data-center operator who is not in the futures market does not fall under the CFTC merely because other people are writing contracts on compute. The CFTC has

authority over futures and swaps, and none over the cash market. Spot capacity can be sold to anyone. If the operator opts into being a market, the market is regulated. If the operator only hedges, they can be an eligible contract participant for swaps or an ordinary futures user, and that still does not make them a regulated entity.

Decentralized networks in which households sell edge-device compute look like markets and are not the same as a standardized commodity market. If what is sold is spot, it is outside CFTC authority. If it is a future, it is inside. If it is a swap, it is inside. A forward that is not a future is probably outside. These networks have existed for about five years inside jurisdictions that already regulate them under other rules, and commodity oversight of compute is what is new. Whether that oversight is retroactive or whether existing venues are grandfathered was raised and answered later in the session.

On-chain as a definitional requirement. The report's definition was questioned for seeming to require an on-chain market. The suggestion was to widen the label so that off-chain commodity markets are included, because most commodity markets are off-chain. The written framework uses "on-chain"; the discussion's proposed extension to off-chain commodity markets remains open. Settlement still has to be strong enough that the futures price tracks the economic exposure. That is why index inputs matter to the CFTC. Requiring regulatory recognition inside the definition risks circularity if regulators use this work as guidance on how to define the market. In the US the future has to be recognized; the spot contract does not.

The short paper's claim, and a disagreement with its rhetoric. A short paper treats compute as a commodity. The programme links the draft study report Compute-as-Commodity: Observable Flows in Decentralised Compute Infrastructure. This resolves the document reference; its draft status is retained. One view accepted on-chain compute as a substantial opportunity for DeFi and crypto but disputed the claim that it would become the most precious commodity in the world. Compute was placed one level of abstraction above energy, energy above oil, and oil probably above labor. The highest abstraction was called the shakiest. The topic is still worth diagnosing.

Scale figures, all qualified as inexact. Hyperscaler capex guidance in a range of $200 to $725 billion. Cloud share above 60 percent for AWS, Azure, and Google, plus a further 20 percent in the same sentence that was not clearly allocated. H100 rental prices up more than 100 percent since January 2026, with a guess of another 50 to 100 percent over one or two years, possibly on the same hardware as the price of the resource rises. Decentralized-compute market capitalization was not treated as exact: many small networks, and some figures are hidden because contracts are private and a large part of the business is off-chain. On-chain revenue of about $72 million was given as a low estimate. Some reports put Aethir above $80 million a year. The figures are large enough that regulators will care, and small inside crypto, which is itself a niche.

The last twelve months. Benchmarks, futures, and regulatory attention have appeared at the same time as decentralized compute networks have changed emissions, grown revenue, and signed clients onto more flexible purchases, including shorter timelines and 24/7 access. That flexibility is what lets more operators and more investors participate.

Compute resembles electricity more than oil. Compute cannot be stored. Contracts have to

stand in for inventory. Oil is a physical good. Electricity is not. The path from "electricity is a commodity" to a working electricity market was recalled as on the order of 20 or 30 years, much longer than compute's path, which raises the risk of rushing regulation and stifling the market. Electricity market design has a record of serious mistakes. Standardization is harder for compute than for electrons. Specialized inference hardware, an ASIC path rendered as a Groq-style queue [unverified: "rock with a queue"], can be good at inference and not at training. Workload changes how useful a given chip is. Oil has grades; a GPU-hour is a poor standard unit; a CPU-hour is a poor standard unit; the use case changes the effectiveness. Storage is a bad substitute for the contract, because the capacity is either used or not, and the opportunity cost is awkward to book. If the AI build-out reverses, it is open whether compute, and the electricity and fuel under it, become less valuable or whether the market has only shifted. The capital stock deteriorates quickly. It is open whether a future can be written on the maturity of the hardware, and whether principal rights should be split from rights to the remaining service life.

The sequence is backwards relative to gold. Gold was regulated first and tokenized later. Compute is trading on-chain before the regulatory category is settled. The smart contracts already deployed cannot simply be removed. If a US venue that already tokenized compute is later treated as a commodities market, the open operational questions are grandfathering, engagement, shutdown, and prohibition on use, and whether those venues even fit the commodity tests.

Three shapes of decentralized compute. 1. A marketplace. A job, meaning a workload, is posted. Providers bid. A contract holds escrow and settles after the compute is delivered.

2. Subnet-based. Many subnets, Bittensor as the example, each with its own commodity, funded by one base token. Bittensor was described as heavily self-subsidized.

3. Workload-specific. The common, more profitable, and more economically coherent shape: one kind of input. Aethir was the example, with AI compute and cloud-gaming compute. Edge devices and small data centers earn emissions. Emissions were described as fixed at about 1.25 million tokens a day, regardless of how many operators are in the network. Gaming was described as several times more profitable than the other workload, with capacity shifting over time.

Marketplace flow: bids, escrow, on-chain settlement, a fee that feeds the token. Take rates of about 4 to 20 percent were stated, alongside heavy annual token issuance that is not matched by revenue. Client counts, private contracts, and off-chain settlement are not visible. Emissions and some settlement are visible on-chain. Models of the business are unreliable where the private book is hidden.

Four networks. Akash, Bittensor, Render, and Aethir. Aethir was noted as having an office in Japan. Most of these four were expected to still exist in five to ten years if the economic model is sound. The model as diagrammed: emissions leave a treasury and pay operators of the hardware; customers pay usage fees to the organizing network; tokens are burned; emissions increase supply and return value to the treasury. The distinctive claim is that the customers are outside crypto. Most crypto networks were described as golden handcuffs around already-crypto users. Selling compute to non-crypto companies is the exception.

What changed after the short paper. Networks have changed token economics, headcount, and provider mix, and have cut emissions, probably because emissions were too generous. Regulators have noticed compute. The changes were not attributed to regulation; the view was that these industries do not organize themselves around the rulebook. Bittensor was given as about $307 million a year in subsidies against about $17 million in revenue. A later remark used a much smaller annual token count against that same revenue figure; the two emission numbers were not reconciled [unverified]. No compute derivatives existed before the paper. The CME H100 and B200 contracts are the change, with the October start already missed and the November CFTC deadline ahead.

Why supervisors have a blind spot. Three gaps: concentration, visibility, and classification.

Hyperscaler concentration is a stability question that resilience and outsourcing rules are meant to answer. The Big Three hold more than 60 percent of global cloud share, on negotiated contracts and discounts. A figure of about $10 billion was used for the crypto-side compute sector and called large for individuals and small for global financial stability. On-chain operators can leave and move their hardware toward better returns. A data center's hardware is fixed and harder to reposition.

On-chain networks publish a large part of settlement. Cloud contracts stay private. Visibility is therefore an invitation for a supervisor, including the CFTC, to talk to networks that may be offering a regulated service to US persons, earlier than the supervisor would find a private cloud book.

No commodity regulator recognizes these networks. They are a small subsector. Without recognition there is no classification and nothing to bring them under. Aethir was described as operating in 93 countries, including sanctioned jurisdictions, and as largely ignored by the large regulators. Pseudonymous identities remain a general problem for the sector.

Tests the short paper applies. Four gates for whether a chain is doing the work of a commodity market: settlement, identity, slashing, or composability that actually needs on-chain coordination. Some real-world-asset chains, Plume as the example, and Aethir operating somewhat like a layer 1, are scaling a service more than they are running an exchange-like market. It is open whether the network or the organization operating it would fall under the guidance.

The six definitional dimensions scored informally:

Akash: fungibility only partial, because the hardware behind it is mixed; price discovery covered; settlement covered; storage unsolved, pending how the CFTC and the CME treat the storage element of compute; roles covered; recognition none.

Render was called the closest on recognition. Aethir might move closer if it standardizes the hardware it adds.

A basket of phones and PCs is not an H100 or a B200. Networks that offer only those standard GPUs are closer to the futures contracts than edge-device networks are.

Observable flows: public and reproducible from blocks, time-stamped, and without KYC. Bitcoin mining pools now generally KYC, and that was called an industry standard. Compute providers here are doing something analogous to mining and are not KYC'd. Who is providing the workload is weakly known. Whether anyone can check that without permission was left as a design question.

Productive versus circular: real demand has to be separated from tokens recycling inside the protocol. External counterparties, real output, and stablecoin settlement are the tells. Circular flows are the warning.

Aethir was credited with enterprise buyers. Akash with machine-learning users. Render with real rendered frames. Inference responses were credited to Akash. Akash and Render settle at least partly in stablecoins. Bittensor subnets were placed in a middle, more circular tier, with some external users.

A drop-checklist for a regulator using only public data, without the operator's help:

If operators withdraw, does the network still function? The answer given was no.

Are there settled prices for one comparable unit across providers? Hard unless the offering is standardized on something like an H100 or a B200.

Is any settlement or arbitration off-chain? Sometimes the chain ratifies a result and the economics are still opaque. Business-development contracts, Aethir was the example, are hidden from the network. How much of the emission stream is actually funded by clients cannot be read off the chain.

What would have to change for a formal designation to be possible? Not answered.

What share of volume ends in non-protocol economic activity? Not measurable without the operator.

The summary score in the captions — substrate plus four of six criteria, and productive activity that is also circular — is too garbled to treat as a finding. The component scores above are what was actually said.

Is compute a commodity the way electricity became one, and who hedges? A statement attributed to CFTC leadership, that America cannot win the AI race without a robust market for compute, was read as a heavy claim and not further sourced. The unit question and the hedger question were taken up directly.

A basket of edge devices does not become one commodity at a single threshold. Commodities already come in grades. US wheat has three futures that the trade often says are not substitutes, because they feed different baking and feed uses. The CME's two chip contracts are an attempt to standardize two grades. A great deal of compute is already traded, including Bitcoin hashing, over the counter, in decentralized bilateral markets. Most physical commodities are traded that way.

Commodity and tradable are different. Fungibility makes the commodity: one electron is like another. A market of buyers, sellers, and intermediaries makes it tradable. Early electricity was a central station selling to local customers. Because electricity is hard to store and hard to transport, the result was local monopoly. Late in the twentieth century, once a transmission grid and multiple generators existed, regulators pushed markets into existence so interconnection could capture scale. Interconnection was administered centrally first and turned into a market later. California in 2000 failed because shortage costs are extreme: if generation cannot meet demand, the grid can collapse, and the spot price can rise without a practical cap. Texas later showed a similar pattern. Compute does not have that shortage cost. Unused compute does not collapse a grid. It is a service. If it is not used, it is not used, although idle hardware still has electricity and maintenance cost.

On that electricity timeline, compute is early and moving fast, inside an AI boom. Announced data centers imply electricity demand beyond what can be produced soon. How that power is obtained is unresolved. Demand is coming from hyperscalers and from labs. A market already exists. Software can move a job between data centers when the hardware is generic. There is always a baseline of maintenance. Prices will swing. Idle capacity earns whatever incremental revenue it can, so overcapacity drives the price toward zero. Shortage plus AI demand drives it toward whatever buyers will pay. That volatility is the case for a market: it balances supply and demand and tells builders what the service is worth.

Hardware that burns out or is obsoleted every product cycle does not, by itself, destroy the commodity. A megawatt-hour is still a megawatt-hour after the generator is scrap. Demand will remain for compute hours on a named chip, or for a flop-based unit, whichever the contract uses. The CME pattern is expected to be: two live generations now, and a new contract when the next generation ships. Activity in futures is concentrated in the near term. Agricultural contracts sometimes reach the next harvest. Listings five and ten years out are typically thin. The value of a given generation's contract should decline as that hardware declines.

Physical delivery is the dangerous design. A short that promises hardware it cannot deliver is in trouble. The CME contracts avoid that by settling to an index, so the question becomes what the index does when Blackwell-class hardware is obsolete. The CFTC was urged to require a refresh procedure before approval. The expected mechanism is a roll into the next contract, translated through price: if old hardware earns $2 per GPU-hour and new hardware is five times as effective and earns $10, a five-for-one adjustment folds the old contract into the new one. Stock-option adjustments after a split were the analogy. That part was called a fairly easy design problem.

Whether anyone actually needs the hedge. The challenge: large AI companies can sign long-term contracts with hyperscalers, so the natural long hedger may not exist. Sellers of the neo-cloud type are a US phenomenon. Japan does not have that seller base. Liquidity may be insufficient for the contract to function. The natural hedger was not identified to the satisfaction of the question.

The electricity analogy was used again. At the start of generation deregulation, large utilities could move the price by withholding plant. That oligopolistic behavior was a large part of the California 2000 breakdown. The compute question is whether competition is now broad enough. The market was argued to be global: training in Iceland or South Africa can be equivalent, except where data have to stay in a known place.

A second hedging story does not require today's hyperscalers to hedge. New industries consolidate. Most of the firms consuming compute now may not exist in ten or fifteen years, as in the dot-com period and earlier industrial booms. If a large share of today's demand sits outside the durable top tier — figures of 60 percent and 90 percent were both used illustratively — an asset manager can short compute on the view that the forward price is too high. That is different from electricity, which keeps houses and buildings running after a technology cycle. A crash in the compute future is possible and is one of the things futures are for: people put money on the forecast. Other downside paths named: a more efficient algorithm, better chips. On the upside constraint, announced data-center power is

not buildable on the timetable markets seem to be pricing. A three-gigawatt chip deal was compared to three nuclear plants of Chernobyl scale for one site. Plants, gigawatt-scale generators, and large transformers have long lead times and are not shelf goods. Markets may be pricing the power years before the infrastructure exists.

Hurdles on the way into regulated markets. Regulatory complexity. The rules exist for reasons. A move-fast culture discovers a morass when it does not know how to build a compliant entity or how regulators work, especially if the service may already be a regulated one. The spot market for compute was not expected to be regulated like the stock market. Farmers face rules on fertilizer and runoff and do not thereby become financial entities. Selling clock cycles from a data center is not a natural job for a financial regulator. Running a futures exchange inside that data center is.

The index and manipulation. An index is hard to manipulate when the data are many publicly visible transactions, and easy to manipulate when a few people state a number. LIBOR was the caution. It was a survey of a handful of London banks, it became the base for hundreds of trillions of dollars of swaps, loans, and bonds, and traders colluded to push the fix up or down depending on their book. A monthly average, which is what the CME construct uses, forces a manipulator to move the market for the whole month. Who sets the price when most capacity is still sold under older private contracts was asked and not really answered.

On-chain prices as a sample of the whole market. If on-chain compute is about $10 billion and the off-chain market is ten or a hundred times larger, public chain prices matter only if they are a representative and statistically large sample of the population, not if they are a complete census. The view in the session was that the on-chain sample can be economically meaningful. That comment is to be written into the paper.

Compute futures and the equity market. The question, tied to the previous day's moves in AI-infrastructure equities, was whether a compute market arrives in time to smooth the cost of capital and extend the AI rally, or whether it adds leverage to a debt cycle that is already too hot, and whether the net effect on equities is neutral. The answer: the link is informational. The AI boom has the shape of a bubble to many observers. New industries show entry, then competition, then consolidation. Anything that feeds attention can keep a bubble inflated. A skyrocketing compute future would tell the equity market that compute demand, and therefore chip makers, are more valuable. A collapsing compute future would tell a chip company and a data-center developer the opposite, and would be a reason not to build. The social benefit claimed is better information about future demand. A high futures price alongside a business that cannot actually sell to customers would be a confusing signal. Prediction markets were noted as another way to produce the same information; little stops a venue from self-certifying a contract on whether the H100 GPU-hour price is above or below a level at a named future time. Markets both produce and consume information.

Who the paper is for. Three audiences were named. The crypto industry, so it sees that traditional commodity regulation is arriving for compute. Regulators, so they see that this industry already exists inside crypto, and so the paper can help define what a compute commodity is. The decentralized-compute operators themselves, so they understand the market they are already in, a book described

again as on the order of $10 billion, and so they are not surprised if someone asks why they are running a market that may be unregulated. One use of the paper is diagnostic: an operator might set out, in a legal white paper, why it is not a commodity and why it is not running a market. Academic work on compute-as-quantity was said to be scarce. BGIN was described as early, and as able to speak both to crypto and to the AI industry. The paper is also meant to travel through BGIN to US market regulators, to prospective new entrants, and to readers in other jurisdictions. What change in reader behavior is expected was asked and answered only at that audience level.

Does a market need a chain? Conventional exchanges have run markets for thousands of years, and derivative markets go back at least to the Osaka rice market. A chain is a database with useful properties, including smart contracts, and it can be more efficient. It is not the only possible technology. The conclusion was to let venues compete and see which technology organizes the market better.

Emissions as subsidy, working fluid, or distortion. Fixed emissions pay operators. If clients do not cover that cost, the emission is a subsidy, and it can function as a marketing cost or as a price distortion. The industry argument for emissions is chicken-and-egg: without a reward, nobody operates hardware, and then nobody pays. That was called a slippery slope. Tokens were described as working fluid, necessary for the mechanism and valuable only up to a point, with the supply set by the promoters. That supply decision is the tricky part. Marketing itself is a normal business function: product, information, distribution. The test is whether new tokens are retained by promoters or used to grow the network. A network that depends on the subsidy for too long will not price its service. A smart contract that emits a fixed amount every year is not pricing to the market. The numerical illustration was inconsistent with the earlier Bittensor figures and is left unverified: a high token price, about $17 million of annual revenue, and a stated annual emission count that does not match the earlier "1.25 million tokens a day" or the "$307 million subsidy." If clients pay a million dollars, the open design question is whether the network emits about a million dollars of rewards or a hundred million. The short paper's emphasis on real economic output was endorsed. Failed crypto experiments are an acceptable cost of experimentation. The surviving test is goods or services that people outside the system will pay for.

Venues that traded before the rule existed. The preferred approach was amnesty for honest operations that are not an obvious scam: people trying to do the thing, not to defraud. Regulation should start from the problem and the best way to solve it. Enforcing a rule only because it is a rule, including a paperwork-count violation, was called bad regulation. "Same activity, same risk, same regulation" was accepted only if the risk is the specific risk. A different technology changes the risk in small ways. Taxis in New York needed a bulletproof partition because drivers were robbed for cash. A platform ride service that does not carry cash does not have that risk. Both are paid rides. Identical rules are the wrong conclusion. The posture favored was light touch, and grandfathering of existing players, aimed at four outcomes: no fraud; customer funds do not disappear; the activity helps the economy grow; the activity does not create systemic risk of the sort that produces mass unemployment. Those are the reasons financial markets are regulated.

Telling a decentralized compute market's risk profile from a traditional data center's requires under-

standing the market. Regulators are usually a couple of years behind, and the gap does not close. In a fast market the practical method is to wait until a real failure is visible — failed delivery, escrow that falls apart — and then write the rule to that failure. Operators were urged to watch their own activity closely enough that, if a regulator arrives, they can show they tried to find and mitigate the problems. Watchful waiting is the supervisory posture that was recommended. Legitimate players want clear rules and should be listened to. Their advice is self-interested, and a wide enough set of conversations filters the bias. Keeping fraudulent actors out of financial services was described as in every serious participant's interest.

Tokenized first, regulated second, does not require a new theory of regulation. Gold's order was regulated, then tokenized. Compute's order is the reverse. The historical pattern in insurance, banking, securities, and commodities is the second one: the product exists because it is useful, then a failure appears (an insurer fails, a bank is run, a market crashes, onion futures are manipulated), then regulation is written. A few post-communist transitions that tried to create a financial system overnight were the exception. After the first statute, the industry negotiates to keep the burden down. Self-regulation appears where standards are easy to agree. Everyone wants bad actors excluded and prefers that someone else pay, which is why the public sector ends up funding the free-rider problem, and why who pays becomes a political fight.

Next papers. The report set still has to be finished. A horizon of about three to four weeks was given for publication or at least for another round of comments. Definitions that came up in the session need another pass. Three candidate papers, plus a filter for choosing among them:

1. Retrofitting. Venues already live on-chain, and how to fold them in, grandfather them, or leave them unbound when a regulator approaches.

2. Benchmark integrity. Whether public on-chain sale prices can feed or cross-check compute price indexes. This requires a standard first. A random hardware basket cannot be priced as an H100 without a conversion. One canonical compute unit, or a published conversion among GPU-hours, tokens, and flops.

3. The token-versus-service perimeter. Where the token is supervised, where the compute service is supervised, and by whom.

The selection rule offered: do the greatest good with the resources available, and name the audience. Two audiences dominate. Industry that is building the markets, and potential regulators. For regulators the useful document educates, describes the regulatory risks, and offers ways to regulate an industry that already exists. People in the room with more regulatory experience were asked to say which of those products would actually be used. The unit-conversion paper was called deep in the weeds and of unclear readership beyond specialists. The tokenomics paper is interesting, and needs a message beyond documentation, though documentation alone was called valuable. Target the environment before starting. Unused prepared questions are to be sent individually. A study report of this session is to be circulated for correction.

The afternoon session in this room was flagged as a stablecoin policy outlook across jurisdictions,

including gaps and MiCA 2.0, after a break until about 14:45, running until about 16:00.

Matters not resolved

Whether compute is "the most precious commodity." The short paper's rhetoric was resisted: compute sits at a high, shaky layer of abstraction above energy, oil, and labor, and is still important.

Whether the commodity-market definition should require an on-chain venue. Widening it to ordinary off-chain commodity markets was proposed. Whether regulatory recognition belongs inside the definition, given circularity, is open.

Self-certification versus the full CFTC request. These contracts were said to have taken the full path. Start dates conflict: 5 October already missed; automatic approval on 9 November; a hope of trading before 5 November.

Whether a data-center spot seller, or a household edge-device seller, is in a commodity market. Spot is outside the CFTC. Futures and swaps are inside. Forwards that are not futures are probably outside. Grandfathering versus retroactivity was answered later in favor of grandfathering honest venues, and was not adopted as a group decision.

Index integrity. Private rental prints may not reflect the market. LIBOR is the failure mode. A monthly average helps and does not finish the question. Who sets the price while most capacity still sits under old bilateral contracts was not answered.

Standardization. A GPU-hour, a CPU-hour, a flop, and a basket of edge devices are not the same commodity. Wheat-style multiple grades are the expected outcome. Networks limited to H100 or B200 hardware are closer to the CME contracts than general edge networks.

Storage and inventory. Compute cannot be stored. How the CFTC and the CME treat storage inside the definition is unresolved. So is a future on hardware maturity, and a split between principal rights and remaining service life.

Whether electricity's history says compute markets are early or late. The journey was called very fast. Shortage-cost dynamics of electricity were said not to apply. Idle-capacity pricing toward zero, and shortage pricing toward whatever buyers will pay, were both asserted.

Who the natural hedger is. Long-term hyperscaler contracts may remove the corporate hedger. Neo-cloud sellers may be a US-only supply. Japan may have no such sellers. Liquidity was questioned. The counter-hypothesis is a global market plus speculative shorts against a cohort of AI users that will not survive consolidation, and against power infrastructure that cannot be built on the market's timetable.

Effect on equities: smoothing the cost of capital and extending the rally, versus adding leverage to a debt cycle. The answer given was "informational, direction unknown," not a forecast.

On-chain revenue, market cap, subsidies, and emission rates. $72 million, "above $80 million," "about $10 billion," "$307 million subsidies versus $17 million revenue," "1.25 million tokens a day," and a later incompatible token count were all spoken. They are not a reconciled set.

Productive versus circular. Enterprise buyers, real rendered output, and stablecoin settlement were offered as evidence of real demand. Hidden off-chain client contracts make the client-funded share of

emissions unknowable from public data. Bittensor was the extreme subsidy case and was still described as having some external users.

Whether a chain is necessary. Useful, not unique. Let the technology compete.

Whether fixed emissions can ever be compatible with a market price. Chicken-and-egg was acknowledged. Long-run dependence on the subsidy was called a failure to price.

How to treat venues that are already live. Light-touch grandfathering and amnesty for non-fraudulent operators, versus "same activity, same regulation" applied generically. Specific risk was the agreed test and was not applied in detail to compute versus traditional data centers.

Wait for a visible failure, versus write rules now. Watchful waiting was recommended. Legitimate firms wanting clarity was also stated. Those two are in tension and were not resolved.

Which next paper: retrofitting, benchmark integrity and a canonical unit, or the token-versus-service perimeter. All three were called good questions. Audience and usefulness are the filter. The unit paper was the one most likely to be too narrow.

"Mature blockchain" as a statutory definition was named and deliberately not discussed.

Disposition of the work

A short BGIN paper on compute as a commodity, covering centralized and decentralized compute markets, is already out. No title was spoken. A longer companion report is unfinished. The report set was described as about three to four weeks from publication or from a further comment round. Session notes and a study report are still to be posted and circulated. Unused questions are to be sent one-to-one. Definitions that the discussion exposed need another pass before the next text.

No status was given for a named harmonization project or a named decentralization-metrics project.

What this session contributes to measurement: the six commodity-market dimensions, the four gates (on-chain settlement, identity, slashing, composability), the productive-versus-circular test, and the public-data drop checklist. They are a diagnostic for compute venues, not a general decentralization score. Partial fungibility, absent recognition, unsolved storage, and hidden off-chain client contracts were the weak cells. A future canonical compute unit, or a conversion among GPU-hours, tokens, and flops, would be a metric if the benchmark-integrity paper is written. It was flagged as possibly too narrow.

What this session contributes to harmonization: the US futures path (CME, CFTC, 45-plus-45 days, no spot authority) does not travel cleanly. Japan was said to lack the neo-cloud seller base that would make a local hedge market. Compute itself was argued to be global except where data residency intervenes. Networks operating in sanctioned jurisdictions are outside any single license. None of that was turned into a cross-border proposal. The token-versus-service perimeter paper would be the harmonization item if it assigns supervisors. Retrofitting is the item that connects this session to the industrial session's "same activity, same risk" line, with the risk required to be specific.

Actions

Each action is written so that it can be tracked without the rest of the record. Where the room named only a person, the action is kept and the name is omitted.

Finish the longer compute report; circulate it for comment on a roughly three-to-four-week horizon; fold in the statistical-sample point and the definitional issues raised in the room.

Post the missing session participation note and the study report.

Choose the next paper by audience. The three candidates are retrofitting of venues already live, benchmark integrity and a canonical compute unit, and the token-versus-service supervisory perimeter. The unit paper needs a reader who is not only a specialist. The tokenomics paper needs a message, not only a description.

For regulators, the useful product is education, a risk description, and a method for an industry that already exists: grandfather honest venues, ignore paperwork-count enforcement, and match rules to the specific risk.

Treat CME index quality as the near-term supervisory fact: where Silicon Data's rental prints come from, what happens when a GPU generation is obsolete, and the November CFTC deadline.

Do not classify edge-device baskets as H100-equivalent without a published conversion. Do not expect the CFTC to regulate spot data-center sales.

Keep watching which networks have external paying customers and stablecoin settlement, and which are emission-subsidized. Bittensor's subsidy-to-revenue gap was the example to keep visible.

Afternoon: stablecoin policy across jurisdictions, gaps, and MiCA 2.0.

Coverage

This record covers the published compute session in Room A. The published Industry Session at 13:00–13:45 is not in the room recording. The next captured Room A session is stablecoin policy after lunch. Figures are reported session claims, not independently validated measurements. Technical names are corrected where supported by the recording, chat, or identified primary documentation. Where a name could not be checked against another part of the same recording, it is left unverified rather than guessed. Logistics, introductions, and names of speakers are omitted.

The slice opens with Japanese small talk, a recording check, and university-teaching chatter, then the session proper. Thanks and audio setup are ignored except here. A personal medical remark after the close is omitted. The market-structure briefing and the network walk were both completed. Questions that were actually discussed: electricity as the precedent and the hedger; hardware obsolescence and contract rolls; whether liquidity and natural hedgers exist, including in Japan; hurdles into regulated industry; index manipulation; the equity-market link; audience for the paper; whether a thin on-chain tape is a valid sample; whether a chain is required; emissions as subsidy; grandfathering and specific risk; whether tokenized-first changes the regulatory playbook. A block of prepared questions was skipped and is to be sent separately. The closing logistics point to a long lunch and an afternoon stablecoin session. One participant joined the question queue late; that exchange is included above as

the equities question and the audience question.

Source clarifications and open checks

Emission arithmetic: about 1.25 million tokens a day, versus a later count on the order of a few thousand a year, versus about $307 million of subsidies and about $17 million of revenue. Not reconciled.

Editorial check: The compute paper supports the reported Aethir footprint of 93 countries and gives its own dated revenue estimate. It does not resolve the other capex, rental, or market totals; do not mix dates or substitute its figures.

November 5 versus November 9 as the trading or approval horizon.

The CFTC-leadership sentence about winning the AI race was paraphrased from a garbled attribution and should be checked against the source before any public use.